SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

(a) Full Citation + Domain + ML Problem

Citation: Wang, J., Li, X., Chen, Y., & Xu, H. (2020). *Multi-Agent Deep Reinforcement Learning for Urban Traffic Light Control in Vehicular Networks*. IEEE Transactions on Vehicular Technology, 69(8), 8242–8254. DOI: https://doi.org/10.1109/TVT.2020.2973972 [(doi.org in Bing)]

Domain / Application Area: Intelligent Transportation Systems (ITS), Smart City Traffic Management.

Specific ML Problem: Optimizing traffic signal control at multiple intersections using multi-agent deep reinforcement learning (DRL) to reduce congestion, waiting time, and queue length.

(b) Objective & Research Gap (150–200 words)

The article aims to design an intelligent traffic signal control system using multi-agent deep reinforcement learning to improve traffic flow in dense urban networks. Traditional traffic control approaches—such as fixed-time scheduling or actuated controllers—cannot adapt effectively to dynamic traffic variations and often optimize each intersection independently. Existing RL-based methods mostly focus on single intersections and fail to capture the interdependence between neighboring junctions.

The authors identify a research gap in coordinated, scalable RL-based traffic control capable of handling high-dimensional traffic states. To address this, they propose a multi-agent DQN framework, where each intersection acts as an autonomous agent that learns optimal signal switching policies based on local observations such as queue length, waiting time, and vehicle density. Deep neural networks are used to approximate Q-values, enabling the agents to learn complex patterns and make decisions that improve global traffic efficiency. The proposed system fills the gap by enabling decentralized yet coordinated learning, demonstrating superior performance over conventional controllers and single-agent RL baselines.

Task 2: Methodology Analysis

(a) ML Technique + Dataset Details

ML Technique Used:

- **Multi-Agent Deep Q-Network (DQN)**

- Each intersection = one RL agent
- **State:** vehicle count per lane, queue length, waiting time, current signal phase
- **Action:** change/maintain signal phase
- **Reward:** negative weighted sum of queue length + waiting time (to minimize congestion)

Dataset / Environment:

- **Source:** SUMO-based microscopic traffic simulator
- **Size:** Thousands of vehicles per episode; multiple episodes under varying traffic loads
- **Features:** lane-wise vehicle density, intersection connectivity, signal phase duration
- **Preparation:** Traffic flows generated using realistic distributions (Poisson arrivals); simulator provides real-time state data to RL agents.

(b) Mapping to CO4 + Strength + Limitation

CO4 Mapping:

- **Reinforcement Learning:** Multi-agent RL, Q-learning, reward design
- **Inductive Learning:** Agents generalize from repeated interactions with the environment

Methodological Strength: The multi-agent design enables decentralized learning while still improving global traffic performance. Deep networks allow handling complex, high-dimensional traffic states.

Methodological Limitation: Training relies heavily on simulated environments. Real-world uncertainties (sensor noise, accidents, unpredictable driver behavior) are not fully captured, which may reduce real-world performance.

Task 3: Results & Critical Evaluation

(a) Key Results + Comparative Table

The authors report significant improvements in traffic flow metrics compared to fixed-time, actuated, and single-agent RL controllers.

Model	Avg. Waiting Time	Avg. Queue Length	Throughput
Fixed-time Control	High	High	Moderate
Actuated Control	Medium	Medium	Moderate–High
Single-agent RL	Medium–High	Medium–High	Moderate
Multi-agent DRL (Proposed)	**Low**	**Low**	**High**

The proposed method reduces waiting time and queue length by **15–30%** compared to baselines (as reported in the article).

(b) Critical Evaluation

Realism of Metrics: The improvements are realistic for simulation-based studies. However, real-world deployment may show reduced gains due to noisy sensors, unpredictable traffic events, and infrastructure limitations.

Potential Biases:

- Traffic patterns used in simulation may not represent rare events (accidents, festivals, sudden surges).
- If training and testing scenarios are similar, the model may overfit to specific traffic distributions.

Additional Experiments Suggested:

- Testing on multiple city layouts (grid, arterial, mixed).
- Adding sensor noise to simulate real-world conditions.

- Comparing with advanced industry systems like SCOOT or SCATS.
- Ablation studies on reward components and state features.

(c) Real-World Applicability & Industry Challenges

Applicability:

- Smart city traffic management
- Intelligent intersections
- Connected vehicle ecosystems
- Urban congestion reduction systems

Challenges in Scaling:

- **Data availability:** Requires reliable sensors (cameras, loop detectors, IoT devices).
- **Compute requirements:** Real-time RL inference across dozens of intersections.
- **Safety:** RL policies must be thoroughly validated to avoid unsafe signal switching.
- **Ethical concerns:** Traffic optimization may unintentionally favor certain neighborhoods.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Linked to CO4 Concepts

1. **Reward design directly shapes RL behavior.**
    - Connects to: *Reinforcement Learning – Reward Functions*
    - The article shows how combining queue length and waiting time guides agents toward globally efficient policies.
2. **Multi-agent RL enables decentralized coordination.**
    - Connects to: *Markov Decision Processes & Multi-agent RL*

    - Each intersection learns independently yet contributes to overall network optimization.

3. **Deep neural networks enhance RL scalability.**
    - Connects to: *Statistical Learning – Function Approximation*
    - DQN allows RL to handle high-dimensional traffic states that traditional tabular RL cannot.

(b) Proposed Extension

Proposed Enhancement: Train the multi-agent DRL system using **real traffic data from Coimbatore/Chennai** combined with simulated noise (missing sensor readings, inaccurate vehicle counts). Compare performance between:

1. Pure simulation training
2. Real-data-driven training with noise models

Justification:

- Real traffic logs are available in many Indian cities.
- Adding noise makes the model robust to real-world uncertainties.
- This experiment bridges the gap between academic RL research and deployable smart city solutions.

Expected Impact:

- More realistic performance evaluation
- Improved robustness
- Higher chances of successful industry deployment